16.3 Building a fault-tolerant web application: Imagery

Before you begin the architecture and design of the fault-tolerant Imagery application, we'll talk briefly about what the application should do. A user should be able to upload an image. This image is then transformed with a sepia filter so that it looks fancy. The user can then view the sepia image. Figure 16.7 shows the process.

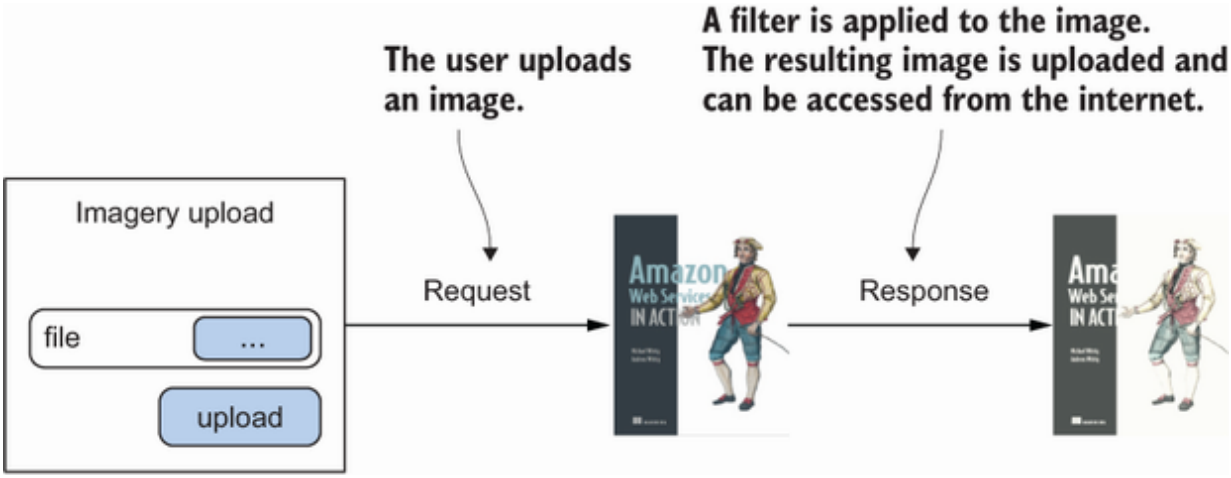


Figure 16.7 The user uploads an image to Imagery, where a filter is applied.

The problem with the process shown in figure 16.7 is that it's synchronous. If the web server crashes during request and response, the user's image won't be processed. Another problem arises when many users want to use the Imagery app: the system becomes busy and may slow down or stop working. Therefore, the process should be turned into an asynchronous one. Chapter 14 introduced the idea of asynchronous decoupling by using an SQS message queue, as shown in figure 16.8.